Cybersecurity Incident Analysis Report: Operation Silent Shadow

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Step 1: Initial Access

1. Overview of Actions of the Attacker:

It was done with the use of spear-phishing attacks with malicious attachments. Innocent workers opened the attachments that triggered embedded malware that was installed backdoor on the computer.

2. Identified TTPs:

T1566.001 Phishing: Spear phishing Attachment

- Malicious attachments delivery to get access.

T1204.002 -User Execution: Malicious File

- Execution of malicious files by users that in turn results in a compromise.

T1059 - Command and Scripting Interpreter

- Distribution of malicious scripts (PowerShell/VBScript) to deploy backdooring.

T1105 - Ingress Tool Transfer

- Use of backdoor to deliver other malicious contents.

Step 2: Perseverance and Privilege escalation

1. Overview of what the attacker did:

Once they got initial access, the attackers got their persistence by placing a persistent service on hacked computers. They also stepped-up privileges through carrying out credential dumping to achieve administrative access.

2. Identified TTPs:

T1547.001 - Boot or Logon AutoStart Execution: Registry Run Keys/ Startup folder

- Achieved persistence that altered startup entries.

T1543.003 - Create/Modify System Process: Windows Service

- Placed malicious services on the computer to provide persistence.

T1003.001 - OS Credential Dumping: LSASS Memory

- Used dumps of passwords to upgrade privileges.

Step 3: Defense Evasion and Credential Access

1. Summary of what the Attacker Acted:

To prevent detection the attackers used obfuscated PowerShell scripts and legitimate tools. They also transferred across the network under stolen credentials

2. Identified TTPs:

T1059.001-PowerShell

- Exploited the use of PowerShell scripts to execute and evade.

T1140 Deobfuscate/Decode Files or Information

– Leveraged obfuscation in order to circumvent detection.

T1021.001 -Remote Services: Remote Desktop Protocol

– Stealing the credentials and moving laterally with them.

T1078 -Valid Accounts

- Was able to use compromised accounts to access laterally.

Step 4 Exfiltration and Impact

1. Summarized Actions Taken by the Attacker:

The attackers stole highly confidential diplomatic information to a malicious command and control (C2) server. Signs of ransomware technique are that they tried to interfere with operations by encrypting important files.

2. Identified TTPs:

T1041-Q Exfiltration Over C2 Channel

– Moved exfiltrated sensitive information via the C2 server.

T1567 - Exfiltration over Web Service

– Possible use of HTTPS to exfiltrate in a stealthy manner.

T1486 - Data is Encrypted to create an Impact

– Staff-logged into the encrypted important files to impede operations.

AG End Result 5: Threat Actor Attribution

1. The APT that has been identified: APT29 (Cozy Bear)

APT29 is the Russian government-sponsored group that specifically targets government ministries, foreign affairs organizations by using spear-phishing attacks as the means of delivery, credential dumping, PowerShell-based evasion, and the use of C2 infrastructure to exfiltrate. The identified TTPs in Operation Silent Shadow are similar to APT29 attributed behaviors.

2. Supporting Analysis:

PT29 has traditionally used malicious attachments in phishing (T1566.001), PowerShell to bypass some security (T1059.001), credential dumping (T1003.001) and the use of C2 to exfiltrate data (T1041). It is highly probable that APT29 is the culprit with the mixture of persistence mechanisms, lateral movement based on valid accounts, and the proliferation of ministries.

Step 6: Mitigation Strategies

1. Prevention Tactics of Initial Access and Execution:

- Implementation of sophisticated phishing protection (sandboxing, file attachment scanning).
- Perform phishing awareness training on a regular basis.
- Block script runs of attachments in email.

2. The following mitigation strategies can be applied regarding persistence and privilege escalation:

- Add registry/service change Endpoint Detection and Response (EDR).
- Activate credential guard and LSASS protection.
- Require and turn over the strong passwords.

3. Defense Evasion and Credential Access Mitigation strategies:

- Activate the extended PowerShell logging and implementation of constraints on execution.
- Make use of multi-factor authentication (MFA) to avoid the lateral use of credentials.
- Install network segmentation to restrict movement.

4. Exfiltration and Impact mitigation Strategies:

- Implement Data Loss Protection (DLP) and observe out-going traffic.
- Institute offline, immutable backup to resist ransomware.
- Detect abnormal data transfer and prevent the malicious connection of C2.

Conclusion

The operation Silent Shadow analysis provided us with an advanced cyber-espionage operation that is attributed to APT29. The threat actors relied on phishing as an entry method, as an access persistence method, dumping credentials, PowerShell in avoiding detection and exiting by exfiltration. The prescribed mitigation fortifies controls

at every stage of the attack chain. A protection-in-depth strategy will allow the Ministry of Foreign Affairs at Eldoria to achieve a much better level of security to safeguard against state-sponsored attacks.